Parmeet Gill

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EDUCATION

University of Manitoba Winnipeg, MB
Bachelor of Science in Mechanical Engineering (Co-op/IIP Option)

EXPERIENCE

Manitoba Hydro – GDD May 2024 – Aug. 2025
Mechanical Engineering Intern Winnipeg, MB

Full-Time: May 2024 – Aug. 2024 | Part-Time: Sept. 2024 – Apr. 2025 | Full-Time: May 2025 – Aug. 2025

- Developed detailed CAD models and assemblies for mechanical components, ensuring precision and compliance with engineering design specifications
- Prepared and revised technical drawings, BOMs, and engineering reports summarizing design updates and site observations
- Conducted Finite Element Analysis (FEA) to evaluate structural integrity, stress distribution, and performance under load conditions for components and mounts
- Performed bolt and joint analysis, verifying safety, reliability, and adherence to engineering standards
- Reviewed technical documentation to verify accuracy and alignment with project requirements, including cross-checking CAD vs. as-built measurements
- Supported installation and commissioning activities, providing engineering guidance to field technicians and reporting findings

Manitoba Hydro – CSD May 2023 – Dec. 2023
Quality Engineering Intern Winnipeg, MB

- Executed Inspection & Test Plans (ITPs), QC record forms, and quality workflow processes in accordance with company standards
- Developed GIS-based field applications, optimized post-processing workflows, and improved spatial data accuracy
- Integrated AI models with Microsoft Power Automate to extract structured data from PDF documents, reducing manual effort and increasing data collection speed
- Designed automated workflows to route extracted data for review, increasing operational accuracy and evaluation speed
- Identified and documented non-conformances, tracked corrective actions, and verified timely closeout to maintain compliance
- Collaborated with cross-functional teams to improve QC documentation and standardize inspection procedures across projects

ENGINEERING SKILLS

Mechanical Engineering & Design: Material selection using engineering calculations, custom part design in SolidWorks, FEA for structural integrity and stress analysis, topology optimization planning, collaborative design development
Biomedical Engineering: MCL-specific orthopedic device design, FMEA for medical devices, regulatory compliance (FDA, Health Canada, ISO 13485/13404/22535), biomechanical testing protocol development, skin-contact material evaluation
Fluid Mechanics & CFD: Pipe flow analysis, pump selection and sizing, nozzle design, SolidWorks Flow Simulation, conservation of mass and momentum applications
Project Management: Coordinated undergraduate engineering design team activities, strong task prioritization, deadline-driven execution
Communication: Technical report writing (design models, cost analysis), collaborative problem-solving, continuous improvement participation

TECHNICAL SKILLS

Engineering Software: SolidWorks (CAD, FEA, Flow Simulation/CFD), Autodesk Inventor, Autodesk Vault, ANSYS
Microsoft Tools: Excel, Word, PowerPoint, Teams, Outlook, Planner
Automation & AI: Microsoft Power Automate (AI Models & Flows)
Certifications: WHMIS, Class 5 Driving License
Languages: English, Hindi, Punjabi

PROJECTS

Automated Robotic Cone Inspection System University of Manitoba | 2025
Client: Northern Blower Inc. Winnipeg, MB

- Designed and validated an automated dimensional inspection system using a Fairino FR10 collaborative robot arm, replacing manual cone measurement processes and maintaining $\pm 1/16$ in tolerance
- Engineered a 76-inch modular inspection table with guide bars, centering blocks, sliding plates, and locking pins to ensure repeatable positioning across all cone variants
- Designed and manufactured a custom robotic arm support mount and end effector integrating a digital dial gauge aligned with the robot's Tool Center Point; performed FEA to validate mount structural integrity and table deflection under load

Water Jetpack Design and Analysis University of Manitoba | 2024
Fluid Mechanics and Applications Winnipeg, MB

- Applied conservation of mass, conservation of momentum, and Bernoulli's equation to analytically determine nozzle exit velocity (23.38 m/s), pump head (33.85 m), and required pump power for a 100 kg rider achieving 6 m elevation
- Performed CFD simulations in SolidWorks Flow Simulation on dual-nozzle jetpack geometry at 90° and 45° nozzle radius angles, validating hand calculations within 4.6% error and confirming outlet velocity uniformity via velocity cut plots
- Calculated major hydraulic losses using the Haaland friction factor approximation (f = 0.0181, Re = 571,963) and minor losses for nozzle cross-section transitions, resulting in a total system head of 40.95 m and pump power of 19.59 kW
- Selected aluminum alloy for pipe and nozzle construction based on lowest surface roughness (e = 0.05 mm), corrosion resistance, and weight reduction, minimizing head losses and prolonging pump lifespan
- Evaluated and selected a Peerless 4AE10 G pump (7.25-inch impeller, 3525 RPM) at 76% efficiency to satisfy the required volumetric flow rate of 589 GPM at 134 ft of total head
- Designed a dual-nozzle swivel joint assembly with machined ball-and-seal bearing interfaces, enabling rider-controlled nozzle rotation for in-flight maneuverability and stability

Redesigned Knee Brace for MCL Injuries University of Manitoba | 2024
Design of Biomechanical Devices Winnipeg, MB

- Co-designed a lightweight MCL-specific knee brace using Nike Flyknit compression material with removable hinged side supports and adjustable Velcro straps, targeting athletes returning to sport after Grade 1 and 2 MCL tears
- Conducted a Failure Modes and Effects Analysis (FMEA) to systematically identify, rank, and mitigate potential failure risks across all brace components using Severity, Occurrence, and Detection criteria
- Outlined plans for stress and strain analysis, cyclic loading simulation, and topology optimization of the hinged side supports and seat belt buckle joint connection to minimize material use while maintaining structural performance
- Assessed regulatory compliance pathways under FDA Class 1 and Health Canada classifications, including ISO 13404, ISO 22535, and ISO 13485 quality management system requirements for market release
- Defined biomechanical, goniometric, impact, durability/fatigue, and skin sensitivity testing protocols for design validation, referencing ISO 10993-10 and ASTM F2808 standards for skin-contact medical devices